

Backtest Summary – Portfolio Rotation Report

Introduction to the Backtesting Methodology

Objective

The purpose of these backtests is to assess the robustness and dynamic consistency of the Portfolio Rotation Report’s analytical engine across different asset classes, observation frequencies, and market structures.

The objective is twofold:

- to verify the reliability of the signals generated by the report under contrasting market conditions (calm, bullish, stressed, rotational, etc.);
- and to validate the stability of the interpretation engine (return, momentum, volatility, regimes).

Each backtest is therefore a simulation “under real-world conditions”: the report’s parameters are applied to an actual dataset downloaded directly from Yahoo Finance.

Data Windows and Observation Intervals

To ensure the reliability, rigor, and transparency of the report-generation engine, a set of **targeted tests** was designed. These tests are not intended to evaluate the model’s performance in terms of returns, but rather to validate the program’s internal consistency, structural robustness, dynamic behavior, and operational readability.

- The goal is not to multiply mechanical checks, but to **select concrete and relevant cases** that reflect the diversity of situations the engine may encounter:
- different asset classes,
 - various analysis frequencies,
 - contrasting market conditions,
 - and data qualities ranging from highly liquid to heavily noisy.

Each test was designed to answer a specific question regarding the engine’s coherence—whether it concerns the reliability of an indicator, the relevance of a dynamic comment, the stability of the calculations, or the clarity of the visual outputs.

Test Category Overview

Category	Description	Associat ed IDs	Dominant Logic
 Internal Consistency (C)	Verifies the accuracy and internal logic of the engine: alignment between generated comments, labels, classification rules, and data structure.	C1 to C14	Validation of the logical structure and internal reasoning of the engine.
 Structural Robustness (R)	Evaluates the engine’s ability to remain stable and coherent when facing imperfect, incomplete, or extreme data.	R1 to R9	Stress testing, edge-case handling, algorithmic resilience.
 Dynamic Behavior (D)	Observes the engine’s responsiveness to context changes (market, frequency, period) and the coherence of dynamic adjustments.	D1 to D5	Verification of response stability and proportionality.
 Readability & Ergonomics (L)	Validates the clarity, fluidity, and visual coherence of the reports, including consistency between tables and charts.	L1 to L3	Verification of visual rendering and user experience.

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List of verification axis

ID	Verification Axis	Specific Objective (engine logic)
C1	Complete sector diversification	Verify that the engine correctly reports full coverage of the 11 GICS sectors ("Full macro coverage" message).
C2	Partial or missing sector coverage	Verify the consistency of "Partial coverage" or "Missing sectors" messages based on absent GICS groups.
C3	Consistent sector biases (growth / defensive / cyclical)	Verify that detected biases (growth-oriented, defensive bias, etc.) match the sectors present.
C4	Market-cap tilt and barbell structure	Check the logic behind "Large/mega-cap tilt" / "Small/micro-cap tilt" / "Barbell by size".
C5	Asset type mix and REIT/ETF/ADR exposure	Verify the correct appearance of comments regarding REIT / ADR / ETF-heavy construction.
C6	Liquidity stance	Verify the correct formulation of messages based on the proportion of illiquid tickers.
C7	Leverage and Beta adjustments	Verify bucket adjustments according to leverage and beta labels ("high", "moderate").
C8	Volatility profile and regimes	Verify the correspondence between volatility-profile labels and "dynamic/speculative" commentary.
C9	Global macro coherence	Verify that the final concatenation provides a coherent, contradiction-free macro view.
C10	Internal logic of short-term scenarios	Verify the full chain "labelling → scenario → fallback" inside <code>_ct_decide_scenario</code> .
C11	Aggregate textual rendering	Verify the formatting of aggregated text: ordering, punctuation, and PDF rendering.
C13	Visual coherence between tables and charts	Verify that values displayed in charts strictly match the data in the tables.
R1	Return (%)	Verify the accuracy of return calculations and their graphical consistency.
R2	Volatility (%) & Volatility Ratio	Verify the consistency of volatility values and ratios across multiple regimes.
R3	Momentum (%) & Momentum Ratio	Verify that momentum variations and ratios follow the actual price dynamics.
R4	Risk-Adjusted Return (RAR)	Check that RAR scores respect the logical hierarchy of Return/Volatility.
R5	Signal Stability	Verify that "stable/volatile" signals are coherent across series with different noise profiles.
R6	Volatility Regime	Verify consistency between regime category ("subdued/normal/elevated/turbulent") and empirical variance.
R7	ADV10USD	Verify that median ADV10USD aligns with displayed liquidity profiles.
R8	Sensitivity to extreme values (outliers / jumps)	Verify that indicators remain stable/reasonable in the presence of jumps (earnings, flash crash).
D1	Beta	Verify that calculated and labeled betas correspond to the asset's behavior relative to the benchmark.
D2	Variation in returns (percentage points)	Verify consistency between raw variation and assigned "strong/weak/neutral" labels.
D3	External adjustments (liquidity, leverage, beta)	Verify bucket adjustments via <code>_ct_adjust_bucket</code> in combined-edge cases.
D4	Extreme correlations	Verify that strong/anti-correlated values are correct and visible on the matrix.
D5	Inter-frequency coherence (Daily / Weekly / Monthly)	Verify that relative trends remain coherent across frequencies.
L1	Number of tickers (global readability)	Verify visual stability and page readability for 10 to 60 tickers.
L2	Stability by window size	Verify indicator coherence for short, normal, and long windows.
L3	Data of low quality (small caps / illiquid)	Verify engine robustness on recent or low-volume tickers.
L4	Final PDF coherence	Verify that the final report remains clean, aligned, and not visually overloaded.
C14	Inter-report consistency	Verify that two identical reports differing by a single variable remain consistent.
R9	Handling of noisy / NaN data	Verify engine stability when facing missing data, outliers, or inconsistent values.

List of Backtest Scenarios Used

Scenario ID	Data Window	Frequen cy	Universe (Number of Tickers)	Main Objective	Associated Tests
S1 — Equity US (Standard)	2020-01-01 → 2024-10-22	Weekly	AAPL, MSFT, NVDA, JPM, JNJ, PG, UNH, XOM, HD, VZ (10)	Verify general CT/LT consistency on large caps	C1, C3, C9, C10, R1, R3, D1, C13
S2 — Equity Europe (Cross-market)	2020-01-01 → 2024-10-22	Weekly	ORA.PA, TTE.PA, AIR.PA, BNP.PA, DG.PA, SAN.PA, SU.PA, AI.PA, RI.PA, MC.PA (10)	Verify engine robustness on European markets	C2, C3, C9, R1, R3, D1, L1, C13
S3 — Short-Term US Stress	2021-12-01 → 2022-06-30	Daily	TSLA, NVDA, META, AMZN, NFLX (5)	Test short-term stability under stress	C7, C8, C10, R2, R5, R8
S4 — Commodities (Cross-class)	2018-01-01 → 2025-10-22	Weekly	XLE, GLD, SLV, DBC, USO, DBA, GSG, IAU (8)	Verify sector-scale factors	C5, R1, R2, R3, R4, D4, C9
S5 — FX Majors (Multi-asset)	2019-01-01 → 2025-10-22	Daily	EURUSD=X, GBPUSD=X, USDJPY=X, AUDUSD=X, USDCAD=X (5)	Verify Return/Momentum consistency for FX	R1, R3, D2, L2, C10, D1
S6 — Crypto (Extreme Volatility)	2020-01-01 → 2024-10-22	Daily	BTC-USD, ETH-USD, SOL-USD, ADA-USD, XRP-USD (5)	Test CT/LT stability under high volatility	C8, C10, R2, R5, R7, R8
S7 — Bonds (Low Yield)	2023-01-01 → 2024-10-22	Weekly	TLT, IEF, SHY, BND, TIP (5)	Verify consistency of weak signals	R1, R2, R3, R6, D3, C9
S8 — Multi-Assets (Cross-class)	2020-01-01 → 2024-10-22	Monthly	SPY, AGG, GLD, DBC, VNQ, XLK, XLF (7)	Verify inter-asset-class consistency	C3, C9, D2, D3, R4, D4, C13
S9 — REITs (Illiquidity)	2013-01-01 → 2024-10-22	Weekly	VNQ, IYR, QQQ, SPY, XLRE (5)	Verify signals on illiquid assets	C5, C6, R3, R7, L3
S10 — Short Sample (Short Window)	2024-06-01 → 2024-10-22	Daily	AAPL, MSFT, NVDA (3)	Test behavior with short historical windows	L2, R1, R3, R4, C10, R5
S11 — Leveraged ETFs	2020-01-01 → 2024-10-22	Daily	SPXL, TQQQ, SOXL, SPXS (4)	Verify amplified behavior	C7, R5, D1, R2, C10
S12 — IPO / Recent Data	2021-10-01 → 2024-10-22	Daily	ABNB, COIN, META, UBER (4)	Verify robustness on recent histories	R1, R8, R9, L2
S13 — Small Caps (High-β / Low ADV)	2023-01-01 → 2024-10-22	Daily	IJR, SLYG, IJT, IJS, PSCT, PSCH, PSCE, VBK (8)	Verify β labels and liquidity behavior	C4, C6, R7, R3, R5, D1
S14 — SMID US+EU (Regime)	2018-01-01 → 2024-10-22	Weekly	IJH, MDY, FNX, IEUS, SCZ, EWUS, IVOO, VO (8)	Verify Momentum / Return / β impact	C8, R1, R3, D2, D1, C10
S15 — Covid Crash (Extreme Volatility)	2019-09-01 → 2020-06-30	Daily	SPY, XLE, QQQ, XLV, XLF (5)	Major stress test on regime shifts	C8, C10, R2, R6, R5, D1
S16 — Inflation Spike (Macro Test)	2021-01-01 → 2023-01-01	Monthly	XLE, XLB, XLF, XLU, SPY (5)	Verify inflation/energy sensitivity	D1, D3, R6, R3, R4
S17 — Cross-Frequencies (Comparison)	2020-01-01 → 2024-10-22	Daily / Weekly / Monthly	SPY, QQQ, XLE, XLF, XLK (5)	Verify inter-frequency coherence	D5, C14, D2, C10
S18 — Extreme Windows	2024-01-01 → 2024-02-01 (15 days) / 2020-01-01 → 2024-10-22 (180 days)	Daily	AAPL, MSFT, TSLA, META (4)	Stress test on window length	L2, R2, R5, C14
S19 — Visualisation / Graph Matching	2022-01-01 → 2024-10-22	Weekly	AAPL, MSFT, NVDA, JPM, JNJ (5)	Verify table-to-chart coherence	C13, C11
S20 — Correlation Matrix Validation	2020-01-01 → 2024-10-22	Weekly	SPY, QQQ, XLE, TLT, GLD, VNQ (6)	Verify extreme-correlation correctness	D4, C13, C9
S21 — Japan Market (Geographic Diversity)	2019-01-01 → 2024-10-22	Weekly	7203.T, 6758.T, 9432.T, 8306.T, 8035.T (5)	Verify structural consistency outside US/EU	C2, C3, C9, R1, D1
S22 — German Market (Small/ Mid)	2022-01-01 → 2025-10-22	Monthly	SAP.DE, DTE.DE, BAYN.DE, LIN.DE, HEN3.DE (5)	Verify robustness on specific European market	C2, C3, C4, R1, R3, R7
S23 — Compressed Volatility (Neutral Regime)	2021-01-01 → 2022-01-01	Daily	SPY, XLP, XLK, XLU, QQQ (5)	Verify logic of “volatility compression” scenario	C8, R6, R2, R5

S24 — Crypto Vol Spike (Mini-window)	2021-05-01 → 2021-07-31	Daily	BTC-USD, ETH-USD, ADA-USD, DOGE-USD (4)	Verify CT reaction under volatility spikes	R2, R5, C10, R8
S25 — Full-Scale Validation (Global Mix)	2020-01-01 → 2024-10-22	Weekly	15 mixed tickers (Equity, FX, Bonds, Crypto)	Verify overall global consistency and stability	C9, R6, D3, R4, L1, L3, C13, R9
S26 — Tri-region	2020-01-01 → 2024-10-22	Weekly	SPY, QQQ, ORA.PA, SAP.DE, 7203.T, 6758.T (6)	Confirm stability of averages and D3 adjustments across regions	D3, R7, R9, C9, L3

Scenario Application

1 - Equity US (Standard)

Data window	2019-12-30 → 2024-10-21
Frequency	Weekly — 252 observations. Sufficient depth: short- and long-term indicators are statistically reliable. Warning: the observation window exceeds the typical interpretability range for this frequency (+384.6% above the recommended upper bound). Results may blend multiple market regimes, reducing economic signal clarity.
Tickers	AAPL, AMZN, GOOGL, HON, JNJ, JPM, LIN, NEE, PG, PLD XOM
Indicators	Short term: 8 weeks volatility, 12 weeks return, 16 weeks momentum Long term: 28 weeks volatility, 30 weeks return, 80 weeks momentum

S1 – Equity US (Standard)

Test	Objective	Observation	Verdict
C1	Verify that the engine correctly reports full coverage of the 11 GICS sectors.	“Full macro coverage: all 11 major economic sectors are present” is included in the text.	VALIDATED
C3	Verify that detected biases (growth-oriented, defensive bias, etc.) match the sectors present.	Despite the exceptional breadth of sectors covered, all biases were correctly detected.	VALIDATED
C9	Verify that the final concatenation produces a coherent, contradiction-free summary.	The engine returns a complete and well-ordered synthesis, integrating relevant biases and structural characteristics without hierarchical or textual inconsistencies.	VALIDATED
C10	Verify the full chain “labelling → scenario → fallback” inside _ct_decide_scenario.	No double trigger (no conflict between conditions). No unrecognized ticker → therefore no fallback error. Scenario dispersion (3 impact categories) is consistent with metrics in Table 2.a.	VALIDATED
R1	Verify the accuracy of return calculations and their graphical consistency.	Test R1 compares the sum of logarithmic returns calculated by the engine with a formula fully independent and based only on prices: $\log(P_t) - \log(P_{t-12})$. The two results match within 10^{-15} , confirming that the engine computes cumulative log-returns correctly and respects the fundamental mathematical identity. The graphical depiction faithfully follows variations in the table.	VALIDATED
R3	Verify that momentum and its ratio follow actual price dynamics.	A Matplotlib model of price (base 100) and momentum, both short- and long-term, confirms that momentum faithfully reflects actual price dynamics. Negative momentum ratios correspond to downturns (e.g., GOOG, JNJ), while recent increases produce a positive ratio (e.g., LNI). The ratio therefore reflects real movement and is consistent with price evolution.	VALIDATED
D1	Verify that calculated and labelled betas correspond to relative benchmark behavior.	Betas calculated by Yahoo Finance on scenario assets are consistent with their nature (mega-cap US) and relative volatility. The classification used in the report (defensive, market, high-beta) relies solely on the beta value, ensuring direct attribution without ambiguity.	VALIDATED
C13	Verify that values shown in the charts strictly match the table data.	Values shown in the three charts strictly match their corresponding data, both in unit and scale. No divergence, no unintended transformation, and no rescaling occur.	VALIDATED

2 - European Equity (Cross-Market)

Data window	2019-12-30 → 2024-10-21
Frequency	Weekly — 252 observations. Sufficient depth: short- and long-term indicators are statistically reliable. Warning: the observation window exceeds the typical interpretability range for this frequency (+384.6% above the recommended upper bound). Results may blend multiple market regimes, reducing economic signal clarity.
Tickers	AI.PA, AIR.PA, BNP.PA, DG.PA, MC.PA, ORA.PA, RI.PA, SAN.PA, SU.PA, TTE.PA
Indicators	Short term: 8 weeks volatility, 12 weeks return, 16 weeks momentum Long term: 28 weeks volatility, 30 weeks return, 80 weeks momentum

S2 – European Equity (Cross-Market)

Test	Objective	Observation	Verdict
C2	Verify the consistency of “Partial coverage” or “Missing sectors” messages based on absent GICS groups.	The “Missing sectors” section correctly identifies the three absent GICS sectors (IT, Utilities, Real Estate). The economic wording used (“no defensive sectors”, “minimal rate-sensitive exposure”, “underweight in technology”) accurately reflects this absence and produces neither incoherence nor false positives. The textual description corresponds exactly to the sector composition.	VALIDATED
C3	Verify that detected biases (growth-oriented, defensive bias, etc.) match the sectors present.	Detected biases are consistent with the sector composition: a dominant presence of Energy, Industrials, Materials, and Consumer Discretionary clearly supports the “cyclical” and “inflation-sensitive” bias. The complete absence of genuinely defensive sectors also supports the diagnosis. No bias contradicts the sectors present.	VALIDATED
C9	Verify that the final concatenation produces a coherent, contradiction-free summary.	The concatenation of textual blocks produces a coherent and structured view of the portfolio. Sector biases, structural properties, and additional comments follow one another without redundancy. The final narrative is homogeneous and faithfully reflects the macro, sector-based, and structural dimensions analyzed.	VALIDATED
R1	Verify the accuracy of return calculations and their graphical consistency.	The analysis confirms that the short-term return calculation is correct and remains economically interpretable regardless of the number of observations, since it only depends on the last 12 weeks. However, derived indicators (Δ Return in percentage points and CT/LT ratios) become economically empty when the long-term observation window spans several years: long-term dynamics are too diluted to serve as a meaningful comparison basis.	VALIDATED
D1	Verify that calculated and labelled betas correspond to relative benchmark behavior.	Betas provided by Yahoo Finance for scenario assets are correctly extracted and faithfully reflected in the report. The Beta_Label classification rule (defensive < 0.80, market $\geq$ 0.80) is respected and matches observed behaviors for all tickers.	VALIDATED
L1	Verify visual stability and readability of pages for 10 to 60 tickers.	Page layout is stable and readable for a 10-ticker universe: no table overlap, sections 2.a and 2.b remain grouped, and charts are accurately aligned. Validation is strong for this universe size.	VALIDATED
C13	Verify that values shown in the charts strictly match the table data.	Values shown in the three sector charts strictly match real data from the KPI table. No divergence, no unintended transformation, no scale error.	VALIDATED

3 - Short-Term US Stress

Data window	2021-12-01 → 2022-06-29
Frequency	Daily — 145 observations. Sufficient depth: short- and long-term indicators are statistically reliable.
Tickers	AMZN, META, NFLX, NVDA, TSLA
Indicators	Short term: 14 days volatility, 29 days return, 43 days momentum Long term: 49 days volatility, 72 days return, 215 days momentum

S3 – Short-Term US Stress

Test	Objective	Observation	Verdict
C7	Verify bucket adjustments based on liquidity, leverage, and beta labels (“high”, “moderate”).	The analysis confirms that bucket adjustments based on liquidity, leverage, and especially beta are fully consistent in this scenario. The four very-high-beta tickers (AMZN, META, NVDA, TSLA) logically shift into the more severe buckets (“Gradual decline”), reflecting strong sensitivity to observed stress conditions. NFLX, whose momentum shifts in a clear direction, is correctly classified in “Range/Noise”.	VALIDATED
C8	Verify consistency between volatility-profile labels and “dynamic/speculative” comments.	Comparison between the defined classification thresholds in VOL_THRESHOLDS and actual volatility values for the long-term window confirms that the engine correctly assigns the “dynamic” and “speculative” labels.	VALIDATED
C10	Verify the full chain “labelling → scenario → fallback” in _ct_decide_scenario.	The chain functions correctly: return, momentum, and volatility labels are unambiguous; _ct_decide_scenario systematically selects the appropriate scenario based on defined rules. All bucket adjustments (beta, liquidity, leverage) are applied exactly as expected. No ticker falls into an incoherent scenario; fallback logic is aligned with its economic profile. Test validated.	VALIDATED
R2	Verify consistency of CT/LT values and ratios across multiple volatility regimes.	CT/LT values (return, volatility, momentum) shown in the tables exactly match the metrics calculated in the test, with no lag or distortion. Ratios comply with internal rules and faithfully reflect underlying price dynamics. Volatility profiles, beta labels, and regime classifications all match source data. Overall, the calculation pipeline is entirely stable and interpretable, even under strong volatility.	VALIDATED
R5	Verify that “stable/volatile” signals are consistent across series with different noise levels.	The Signal Stability evaluation outputs a “neutral” label, consistent with the highly noisy and non-directional behavior of the tested series. The engine returns a correct signal structure without forcing incorrect interpretations. However, since CT/LT dynamics cancel each other out in this scenario, the signal provides no actionable information.	VALIDATED
R8	Verify that indicators remain stable/reasonable during jump events (earnings, flash crash).	Several extreme price jumps are present in this dataset (notably on AMZN, META, TSLA, and NFLX). The engine correctly absorbs shocks: short-term indicators (return, momentum, local volatility) react as expected, while structural properties (volatility profile, regime, beta, illiquidity indicators) remain stable and coherent. This shows that the algorithm distinguishes short-term noise from underlying asset behavior, strengthening robustness against outliers. KPIs remain interpretable across scenarios, confirming that the logic is sound even under severe disturbances.	VALIDATED

4 - Commodities (Cross-class)

Data window	2018-01-01 → 2025-10-22
Frequency	Weekly — 408 observations. Sufficient depth: short- and long-term indicators are statistically reliable. Warning: the observation window exceeds the typical interpretability range for this frequency (+684.6% above the recommended upper bound). Results may blend multiple market regimes, reducing economic signal clarity.
Tickers	DBA, DBC, GLD, GSG, IAU, SLV, USO, XLE
Indicators	Short term: 8 weeks volatility, 12 weeks return, 16 weeks momentum Long term: 28 weeks volatility, 30 weeks return, 80 weeks momentum

S4 – Commodities (Cross-class)

Test	Objective	Observation	Verdict
C5	Verify the correct appearance of REIT / ADR / ETF-heavy construction comments.	The structure-detection engine correctly identifies asset nature. In this scenario, the portfolio is composed of 100% ETFs, and the dynamic comment “Concentration risk: portfolio heavily exposed to ETF instruments” appears as expected.	VALIDATED
R1	Verify the accuracy of return calculations and their graphical consistency.	All three assets show perfect coherence between price dynamics and short-term return. Accelerations, consolidations, and pullbacks are correctly reflected in return curves without lag. Even in the extreme case (USO), return reaction remains fully logical. The return calculation is validated for this scenario.	VALIDATED
R2	Verify the consistency of CT/LT values and ratios across multiple volatility regimes.	Short-term and long-term volatility values are coherent across the portfolio. Ratios remain stable (0.7–2.2) and no numerical artifact appears, showing that the pipeline works correctly. Regimes match ratios for each ticker. Overall the test validates volatility robustness in varied market environments.	VALIDATED
R3	Verify that momentum and its ratio follow actual price dynamics.	The mathematical and logical checks confirm correct behavior: price direction aligns with short-term momentum, long-term momentum is consistent with structural trend, short/long-term momentum comparison via the Momentum Ratio is coherent, and signals remain continuous. The test confirms accuracy without needing new visual charts.	VALIDATED
R4	Verify that RAR scores respect the logical Return/Volatility hierarchy.	Risk-Adjusted Return rankings follow expected logic: higher return increases RAR, higher volatility decreases it, and negative-return assets systematically produce negative RAR.	VALIDATED
D4	Verify that strong/anti correlations are correct and visible in the matrix.	The most correlated and most anti-correlated pairs appear correctly in the tables, and correlation intensity matches the visual matrix. No spurious or insignificant correlation is present. All displayed values are mathematically and visually correct.	VALIDATED
C9	Verify that the final concatenation produces a coherent, contradiction-free summary.	The final concatenation yields a coherent global view with no contradictions.	VALIDATED

5 - FX Majors (Multi-Asset)

Data window	2019-01-01 → 2025-10-21
Frequency	Daily — 1772 observations. Sufficient depth: short- and long-term indicators are statistically reliable. Warning: the observation window exceeds the typical interpretability range for this frequency (+786.0% above the recommended upper bound). Results may blend multiple market regimes, reducing economic signal clarity.
Tickers	EURUSD=X, GBPUSD=X, USDJPY=X, AUDUSD=X, USDCAD=X
Indicators	Short term: 40 days volatility, 60 days return, 80 days momentum Long term: 140 days volatility, 150 days return, 400 days momentum

S5 - FX Majors (Multi-Asset)

Test	Objective	Observation	Verdict
R1	Verify the accuracy of return calculations and their graphical consistency.	The comparison between indexed prices and CT (60-day) returns for EURUSD, AUDUSD, and USDCAD shows perfect coherence: near-zero returns observed in the table directly result from the low directional amplitude of the Forex market.	VALIDATED
R3	Verify that momentum and its ratio follow actual price dynamics.	Long-term momentum dynamics correctly follow price trajectories for Forex pairs. Pullbacks, phases of consolidation, and accelerations are all properly captured. Test R3 is validated.	VALIDATED
D1	Verify that calculated and labelled betas correspond to relative benchmark behavior.	Betas calculated via daily-return covariance (vs benchmark UUP) are very low (between -0.24 and 0.13), which matches the typical macro structure of Forex, historically weakly correlated to the Dollar Index.	VALIDATED
D2	Verify consistency between raw variation and “strong/weak/neutral” labels.	Labels assigned (“neutral”, “bullish”, “accelerating”, etc.) for return, momentum, and short-term coherence strictly follow the intervals defined in the model and correspond to actual observed variations in the data.	VALIDATED
L2	Verify coherence of indicators across short, medium, and long windows.	In this Forex scenario, short-term (40–80 days) and long-term (150–400 days) indicators are mathematically coherent. However, economic meaning of long-term signals is strongly degraded: the observation window exceeds +786% of the upper recommended bound in the report, mixing several market regimes and diluting the LT signal.	VALIDATED
C10	Verify the full chain “labelling → scenario → fallback” inside _ct_decide_scenario.	Tests confirm that all labels (return, momentum, volatility) are correctly generated and passed to the scenario-decision module. Selection and fallback logic operate as expected, even in cases where an asset is absent or has partial data.	VALIDATED

6 - Crypto (Extreme Volatility)

Data window	2020-01-01 → 2024-10-21
Frequency	Daily — 1756 observations. Sufficient depth: short- and long-term indicators are statistically reliable. Warning: the observation window exceeds the typical interpretability range for this frequency (+778.0% above the recommended upper bound). Results may blend multiple market regimes, reducing economic signal clarity.
Tickers	BTC-USD, ETH-USD, SOL-USD, ADA-USD, XRP-USD
Indicators	Short term: 40 days volatility, 60 days return, 80 days momentum Long term: 140 days volatility, 150 days return, 400 days momentum

S6 - Crypto (Extreme Volatility)

Test	Objective	Observation	Verdict
C8	Verify consistency between volatility-profile labels and “dynamic/speculative” comments.	Across all tested lines, V_profile correctly reflects structural volatility level, and comments describe only the regime (V_regime) with no contradictions.	VALIDATED
C10	Verify the full chain “labelling → scenario → fallback” inside _ct_decide_scenario.	Tests confirm that all labels (return, momentum, volatility) are correctly generated and passed to the scenario-decision module. Selection and fallback logic behave as expected, even when an asset is missing or only partially available.	VALIDATED
R2	Verify consistency of CT/LT values and ratios across multiple volatility regimes.	Indicators (CT and LT) were manually compared for several volatile assets to assess stability and ratio coherence. Cross-metric examination shows complete consistency: Volatility Ratios are systematically < 1 when CT is calmer than LT; Momentum Ratios track signal weakening or compression; ΔReturn(pp) correctly reflects CT/LT divergences without anomalies.	VALIDATED
R5	Verify that “stable/volatile” signals are consistent across series with different noise levels.	The model behaves as expected: stability labels, volatility regimes, and synthetic signals remain cautious and avoid over-interpretation. One important nuance: ratios can become extremely high in highly noisy environments; this is not a technical bug but a consequence of ratio structure (small LT base combined with unstable CT amplifies extreme values). These ratios reflect structural disequilibrium rather than incorrect signal direction.	VALIDATED
R7	Verify that ADV10USD median aligns with liquidity profiles.	ADV10_USD volumes are massive, typical of a crypto large-cap universe. very_liquid labels are coherent with usual thresholds (≥ 100M USD).	VALIDATED
R8	Verify that indicators remain stable/reasonable in the presence of jump events (earnings, flash crash).	For this test, several windows containing price jumps and rapid rebounds were selected to observe indicator sensitivity. Short-term return, defined from simple ratio price_end/price_start, naturally smooths intraday movements and provides a cold reading of shocks. Volatility, in contrast, reacts fully to amplitude of variations and correctly reflects unstable or explosive market nature.	VALIDATED

7 — Bonds (Low Yield)

Data window	2023-01-02 → 2024-10-21
Frequency	Weekly — 95 observations. Sufficient depth: short- and long-term indicators are statistically reliable. Warning: the observation window exceeds the typical interpretability range for this frequency (+82.7% above the recommended upper bound). Results may blend multiple market regimes, reducing economic signal clarity.
Tickers	TLT, IEF, SHY, BND, TIP
Indicators	Short term: 8 weeks volatility, 12 weeks return, 16 weeks momentum Long term: 28 weeks volatility, 30 weeks return, 80 weeks momentum

S7 - Bonds (Low Yield)

Test	Objective	Observation	Verdict
R1	Verify consistency between volatility-profile labels and “dynamic/speculative” comments.	The test shows complete coherence: no comment suggests a speculative or dynamic environment when the numbers do not justify it. Volatility labels never contradict textual indicators.	VALIDATED
R2	Verify coherence of CT/LT values and ratios across multiple volatility regimes.	R2 verification was performed by comparing CT/LT ratios (return, momentum, volatility) in a low-volatility bond universe to detect potential structural inconsistencies. All observed ratios remain within expected ranges, matching CT/LT levels of underlying assets. Values align with regimes and comments shown in the report. Although signals are naturally less discriminating in low-volatility universes, no contradiction or anomaly was found.	VALIDATED
R3	Verify that momentum and its ratio follow actual price dynamics.	To validate R3, real bond ETF price movements were compared with long-term and short-term momentum on the same grid. Curves show systematic directional consistency and a coherent temporal offset typical of LT momentum, with gradual reactions during pullbacks. Despite slight residual sensitivity on highly liquid assets, no structural incoherence was observed.	VALIDATED
R6	Verify coherence between volatility regime categories (“subdued/normal/elevated/turbulent”) and empirical variance.	For R6, volatility regimes were compared with CT/LT empirical variance. Across all bond series, volatility levels fall in the lower range of historical distributions, consistent with the “subdued” label in the report. No asset displayed sudden volatility spikes or structural compression that would justify a higher regime.	VALIDATED
D3	Verify bucket corrections via <code>_ct_adjust_bucket</code> under combined test conditions.	To validate this test, multiple scenarios were applied to a homogeneous bond universe (BND, IEF, SHY, TIP, TLT) to observe how liquidity, beta, and leverage conditions modify bucket assignment. Results show that labels are correctly adjusted (compression across all tickers), and the system correctly identifies that the group shares the same context, consistent with model logic. Corrections were always coherent: illiquid or defensive profiles only shift buckets when appropriate, without generating undesired systemic effects.	VALIDATED
C9	Verify that the final concatenation produces a coherent, contradiction-free summary.	After full examination of the final output (KPI → ratios → scenarios → aggregated comments), no textual contradiction or inconsistency was detected.	VALIDATED

8 — Multi-Assets (Cross-class)

Data window	2020-01-01 → 2024-10-2
Frequency	Monthly — 58 observations. Sufficient depth: short- and long-term indicators are statistically reliable.
Tickers	SPY, AGG, GLD, DBC, VNQ, XLK, XLF
Indicators	Short term: 5 months volatility, 9 months return, 12 months momentum Long term: 17 months volatility, 22 months return, 60 months momentum

S8 — Multi-Assets (Cross-class)

Test	Objective	Observation	Verdict
C3	Verify that detected biases (growth-oriented, defensive bias, etc.) match the sectors present.	The sector biases detected by the engine (defensive, market, high-beta, dynamic) faithfully reflect the sectors represented in the portfolio. Bond ETFs appear as defensive, tech appears as high-beta, gold as a refuge asset, and SPY/XLF as market references. No contradictory bias appears given the structural nature of the universe.	VALIDATED
C9	Verify that the final concatenation produces a coherent, contradiction-free summary.	The complete chain “collection → calculations → labelling → scenarios → concatenation” produces a perfectly coherent synthesis with no internal contradictions. Final blocks align with all calculation units and the final representation remains economically reasonable.	VALIDATED
D2	Verify consistency between raw variation and “strong/weak/neutral” labels.	Test D2 confirms that “strong / weak / neutral” labels match observed raw variations. Strong directional assets receive bullish labels, stagnating assets are labeled neutral, and ambiguous signals produce cautious labels. Extreme ratios do not destabilize the engine, which maintains a robust interpretative logic.	VALIDATED
D3	Verify bucket corrections via <code>_ct_adjust_bucket</code> under combined conditions.	Validation of correction rules shows that liquidity-, leverage- and beta-based adjustments all function correctly without contradictions. Each ticker receives a final bucket consistent with its risk characteristics. Rule ordering does not generate undesirable effects. A minor expected limit appears for certain very high-beta assets (e.g., XLK), where stricter reinforcement could be justified; however, the current rule remains internally consistent.	VALIDATED
D4	Verify that strong/anti-correlations are correct and visible in the matrix.	The correlation matrix shows a fully coherent structure: equity ETFs (SPY, XLK, XLF, VNQ) form a correlated cluster; defensive or non-cyclical assets (AGG, GLD, DBC) display weak or near-zero correlations as expected. Cross-asset relations correctly reflect observed dynamics (momentum, volatility, regimes). No contradiction is identified. The correlation logic fully matches the scenario’s cross-asset behavior.	VALIDATED
R4	Verify that RAR scores respect the logical Return/Volatility hierarchy.	Assets with the best returns and contained volatility (e.g., SPY) logically obtain the highest RAR scores. More volatile or lower-performing assets (XLK, VNQ, XLK, AGG, DBC) gradually move downward in the ranking, with no anomaly. This confirms that the RAR mechanism correctly reflects the portfolio’s return/risk balance.	VALIDATED
C13	Verify that values displayed in charts strictly match the table data.	A comparison between chart values (return, volatility and momentum bars) and those in the Key Performance Indicators table shows perfect correspondence. Bar heights match the numerical data; small discrepancies stem only from graphic rounding. No transformation, rescaling or mismatch is observed.	VALIDATED

9 — REITs (Low liquidity)

Data window	2020-01-01 → 2024-10-22
Frequency	Weekly — 252 observations. Sufficient depth: short- and long-term indicators are statistically reliable. Warning: the observation window exceeds the typical interpretability range for this frequency (+384.6% above the recommended upper bound). Results may blend multiple market regimes, reducing economic signal clarity.
Tickers	VNQ, IYR, QQQ, SPY, XLRE
Indicators	Short term: 8 weeks volatility, 12 weeks return, 16 weeks momentum Long term: 28 weeks volatility, 30 weeks return, 80 weeks momentum

S9 - REITs (Low liquidity)

Test	Objective	Observation	Verdict
C5	Verify the correct appearance of REIT / ADR / ETF-heavy construction comments.	Scenario validation confirms that the comment module correctly identifies the strong presence of ETFs in the portfolio's construction. Even if the engine does not categorize REIT ETFs separately (since Yahoo Finance labels them as ETFs), sector information appears correctly in the charts, explicitly showing significant weighting in Real Estate / REIT.	VALIDATED
C6	Verify the correct formulation of messages based on the proportion of illiquid tickers.	The labels illiquid / thin / liquid assigned to each asset correspond to ADV10_USD thresholds by market-cap category. Messages in the report ("under limited liquidity", "with ample liquidity") accurately reflect these labels.	VALIDATED
R3	Verify that momentum and its ratio follow actual price dynamics.	R3 validation confirms that long-term momentum faithfully follows real price dynamics. Uptrends, pullbacks, and acceleration phases are correctly translated without over-smoothing or artificial noise reduction. Signals remain coherent, economically justified, and correctly used under stable conditions.	VALIDATED
R7	Verify that ADV10USD median aligns with liquidity profiles.	R7 verification confirms that liquidity levels (thin / liquid / illiquid) align with ADV10USD thresholds defined in the engine. The portfolio's ADV10USD median (~6.9M USD) sits in the neutral zone, consistent with the absence of extremely illiquid assets. No misclassification or skew was detected.	VALIDATED
L3	Verify the robustness of the engine on recent or low-trading-volume tickers.	L3 was validated on a universe consisting exclusively of established ETFs, without recent or illiquid instruments. The objective was to ensure that the stability of momentum, volatility, and return calculations remains intact in the absence of data irregularities. No structural anomalies or stress-induced distortions were observed. This scenario does not stress-test illiquidity directly but confirms that the engine behaves correctly in a stable and liquid universe.	VALIDATED

10 — Short Sample

Data window	2024-05-31 → 2024-10-21
Frequency	Daily — 99 observations. Short-term indicators: OK . Long-term indicators: Unavailable (insufficient historical depth).
Tickers	AAPL, MSFT, NVDA (3)
Indicators	Short term: 9 days volatility, 19 days return, 29 days momentum Long term: 31 days volatility, 47 days return, 145 days momentum

S10 — Short Sample

Test	Objective	Observation	Verdict
L2	Verify coherence of indicators across short, medium, and long windows.	In a universe with shallow historical depth (99 days), short-term indicators remain coherent and aligned with price dynamics. Long-term indicators are correctly disabled and marked as not usable. No CT/LT conflict appears, confirming robustness of estimation logic in limited-horizon environments.	VALIDATED
R1	Verify consistency between volatility-profile labels and “dynamic/speculative” comments.	Qualitative comments correctly reflect the assigned volatility classes. Dynamic and speculative assets maintain a tone consistent with their risk structure, while the tone shifts appropriately when the volatility regime is “subdued.” No mismatch between assigned labels and narrative was detected.	VALIDATED
R3	Verify that momentum and its ratio follow actual price dynamics.	Graphical validation shows perfect coherence between price trajectories and momentum indicators. AAPL shows stable momentum with moderate upward tendency; MSFT shows weakening momentum consistent with a slight downward drift; NVDA shows an acceleration matching a recent upward impulse. The engine faithfully follows real dynamics with no artificial smoothing or unjustified amplification.	VALIDATED
R4	Verify that RAR scores respect the logical Return/Volatility hierarchy.	RAR hierarchy fully respects expected structure: low-return, low-volatility assets display moderate RAR (AAPL); negative-return assets display negative RAR (MSFT); high-return, high-volatility assets preserve higher RAR without excess (NVDA). No logical inversion or incoherent amplification was detected.	VALIDATED
C10	Verify the full “labelling → scoring → scenario → fallback” chain in _ct_decide_scenario.	The full chain operates correctly: momentum and return labels are correctly interpreted; RET_SCORE and MOM_SCORE activate proper decision branches; fallback only triggers when conditions are unmet (MSFT). Scenarios assigned (vol_compress for AAPL, uptrend for NVDA, range for MSFT) fully match price and momentum structure. No logical drift or inconsistency was found.	VALIDATED
R5	Verify that “stable/volatile” signals are consistent across series with different noise levels.	AAPL and MSFT show very low noise and are correctly labelled stable; NVDA, with higher noise, is correctly flagged volatile without excessive interpretation. No ticker receives an exaggerated volatility label inconsistent with the model. Signals accurately reflect real noise levels.	VALIDATED

11 — Leveraged ETFs

Data window	2025-01-02 → 2025-04-14
Frequency	Daily — 70 observations. Sufficient depth: short- and long-term indicators are statistically reliable.
Tickers	SOXL, SPXL, SPXS, TQQQ
Indicators	Short term: 7 days volatility, 14 days return, 21 days momentum Long term: 17 days volatility, 28 days return, 63 days momentum

S11 — Leveraged ETFs

Test	Objective	Observation	Verdict
C7	Verify bucket adjustments based on leverage and beta labels (“high”, “moderate”).	Tickers were evaluated by first applying leverage effect, which systematically increases bucket severity and mechanically amplifies any short-term scenario. Beta was then used as an additional conditional adjustment factor, reinforcing bucket severity for already risk-classified situations. Leveraged ETFs logically shifted into high-impact buckets, confirming consistency and robustness for test C7.	VALIDATED
R5	Verify that “stable/volatile” signals are consistent across series with different noise levels.	Series were evaluated by comparing short-term vs long-term volatility, confirming a “stable” classification when recent noise contracts and “volatile” when dynamics intensify. CT/LT ratios perfectly matched these observations: SPXL, SPXS, and TQQQ display calmer volatility, while SOXL shows structurally high volatility. Signals are coherent with the true nature of each series, validating test R5.	VALIDATED
D1	Verify that calculated and labelled betas correspond to relative benchmark behavior.	Betas were verified by comparing magnitude and sign to expected behavior relative to the benchmark. Leveraged long ETFs show strongly positive beta, while SPXS logically exhibits a negative beta of similar magnitude. Assigned labels (“very_high-beta”) match measured values perfectly, fully validating test D1.	VALIDATED
R2	Verify the coherence of CT/LT values and ratios across multiple volatility regimes.	Short-term/long-term volatility ratios were compared across several tickers with different noise profiles. Series with lower volatility (SPXL, SPXS, TQQQ) show ratios below 1 and are correctly classified in “subdued” regime, while SOXL, more noisy and near “normal” regime, shows higher ratios. Full consistency across ratios, regimes, and comments validates robustness for test R2.	VALIDATED
C10	Verify the full “labelling → scenario → fallback” chain in <code>_ct_decide_scenario</code> .	The short-term decision chain was verified by checking return, momentum, and volatility labels, scenario selection logic, and fallback activation when needed. Primary scenario selection always aligns with logical criteria; fallback triggers only when expected. No contradictory assignment occurred and no scenario broke the chain’s logic. Structural coherence fully validates test C10.	VALIDATED

12 — IPO/ Recent data

Data window	2025-08-01 → 2025-11-14
Frequency	Daily — 75 observations. Sufficient depth: short- and long-term indicators are statistically reliable.
Tickers	AAPL, MSFT, NVDA (3)
Indicators	Short term: 7 days volatility, 15 days return, 22 days momentum Long term: 17 days volatility, 30 days return, 66 days momentum

S12 — IPO / Recent data

Test	Objective	Observation	Verdict
R1	Verify the accuracy of return calculations and their graphical consistency.	Short-term returns were overlaid on smoothed price series, confirming complete coherence in direction, amplitude, and timing. Positive and negative return variations match accelerations and decelerations visible in price curves with no lag or anomaly. CT return calculations are fully validated for test R1.	VALIDATED
R8	Verify that indicators remain stable/ reasonable in the presence of jump events (earnings, flash crash).	Jump events observed in the series (earnings, isolated spikes) were correctly absorbed: short-term indicators reacted without overshooting, while long-term indicators remained stable. No aberrant scenario was generated, and CT/LT ratios remained coherent despite spikes. The engine shows robust responsiveness to jump events, validating test R8.	VALIDATED
R9	Verify engine stability in the presence of missing data, outliers, or inconsistent values.	Series were tested under partial history, jump events, and ambiguous configurations. The engine prefers deactivating certain indicators (LT “Unavailable”) rather than producing unstable values. Shocks affect short-term indicators only, without corrupting ratios or scenarios, while ambiguous cases default to neutral fallbacks. This cautious handling of missing and inconsistent data confirms full stability for test R9.	VALIDATED
L2	Verify coherence of indicators across short, medium, and long windows.	Short-, medium-, and long-term indicators were compared for AAPL, MSFT, and NVDA. CT signals react to local noise as expected, LT signals follow structural trends, and medium-term indicators sit between the two without anomalies. Window horizons produce consistent measurements adapted to each asset’s behavior, validating test L2.	VALIDATED

13 — Small Caps (High Beta / Low ADV)

Data window	2023-01-03 → 2024-10-21
Frequency	Daily — 453 observations. Sufficient depth: short- and long-term indicators are statistically reliable. Warning: the observation window exceeds the typical interpretability range for this frequency (+126.5% above the recommended upper bound). Results may blend multiple market regimes, reducing economic signal clarity.
Tickers	IJR, IJS, IJT, PSCE, PSCH, PSCT, SLYG, VBK
Indicators	Short term: 40 days volatility, 60 days return, 80 days momentum Long term: 100 days volatility, 120 days return, 240 days momentum

S13 — Small Caps (High Beta / Low ADV)

Test	Objective	Observation	Verdict
C4	Check the logic of “Large/mega-cap tilt”, “Small/micro-cap tilt”, and “Barbell by size”.	Verification of market-cap distribution (large/mid/unknown) shows that no category reaches the thresholds required for triggering a tilt or barbell scenario in the engine’s logic. The report therefore displays no size bias, which is consistent with expected behavior. Minor noise comes from “unknown” tickers, but this does not affect the conclusion.	VALIDATED
C6	Verify correct formulation of messages based on the proportion of illiquid tickers.	The strong dominance of illiquid tickers (~90%) correctly triggers formulations such as “under illiquid trading conditions” at the line level and a global diagnostic oriented toward illiquidity. No contradictory messaging appears. Threshold mechanics operate as expected.	VALIDATED
R7	Verify that ADV10USD median aligns with displayed liquidity profiles.	The ADV10USD median (~0.14M USD) is very low, confirming a highly constrained liquidity environment. Individual liquidity labels (“illiquid”, “thin”) and global comments accurately reflect this market context. No incoherent message is observed.	VALIDATED
R3	Verify that momentum and its ratio follow actual price dynamics.	Direct comparison of price series and LT momentum confirms complete coherence: accelerations, consolidations, and decelerations in price are faithfully reproduced in momentum trends. No structural shift or artifact is observed, even for more volatile tickers.	VALIDATED
R5	Verify that “stable/volatile” signals are consistent across series with different noise levels.	Series analysis shows that low-noise tickers receive compressed or normal volatility labels, while more erratic tickers (PSCE, PSCT) receive speculative or amplifying profiles. No structural anomaly is detected between real series behavior and stability/volatility labels.	VALIDATED
D1	Verify that calculated and labelled betas correspond to relative benchmark behavior.	Comparison between calculated betas, assigned labels, and dynamic behavior confirms strong alignment: industrial tickers behave as expected with moderate beta; sector ETFs like PSCE and PSCT show higher sensitivity consistent with “high-beta” labeling. No classification mismatch is detected.	VALIDATED

14 — SMID US+EU

Data window	2018-01-01 → 2024-10-21
Frequency	Weekly — 356 observations. Sufficient depth: short- and long-term indicators are statistically reliable. Warning: the observation window exceeds the typical interpretability range for this frequency (+584.6% above the recommended upper bound). Results may blend multiple market regimes, reducing economic signal clarity.
Tickers	EWUS, FNX, IEUS, IJH, IVOO, MDY, SCZ, VO
Indicators	Short term: 8 weeks volatility, 12 weeks return, 16 weeks momentum Long term: 20 weeks volatility, 24 weeks return, 48 weeks momentum

S14 — SMID US+EU

Test	Objective	Observation	Verdict
C8	Verify consistency between volatility-profile labels and “dynamic/speculative” comments.	Behavioral comments (momentum/volatility/beta) correctly align each ticker with its dynamic or speculative volatility profile. No false positives or negatives were detected. Comment intensity is consistent with the profile. Minor limitation: highly speculative profiles could be described with slightly more explicit wording if desired.	VALIDATED
R1	Verify the accuracy of return calculations and their graphical consistency.	Short-term return calculations are correct, stable, and graphically coherent with price variations. Curves confirm that the model accurately captures acceleration, slowdown, and market stagnation phases.	VALIDATED
R3	Verify that momentum and its ratio follow actual price dynamics.	Each ticker was checked by overlaying price (base 100) and long-term momentum over 48 points. Momentum correctly tracks phases of rise, correction, and consolidation visible in prices. No abnormal lag or excessive smoothing was observed; only the natural slight delay inherent to momentum appears, consistent with a smoothed signal.	VALIDATED
D2	Verify consistency between raw variation and “strong/weak/neutral” labels.	AvgReturn(%) values were compared against ETF weekly classification thresholds. All values fall within the expected interval (~ -1.5% to +1.5%), which corresponds to a neutral label. Labels generated by the engine correctly reflect this universe, with no ambiguity. Classification logic is applied correctly.	VALIDATED
D1	Verify that calculated and labelled betas correspond to relative benchmark behavior.	Beta values were compared to classification thresholds and price sensitivity relative to the benchmark. All ETFs show beta consistent with their structural profiles (mid-caps ≈ 1 , European small caps < 0.8). “Market” and “defensive” labels correctly reflect this sensitivity.	VALIDATED
C10	Verify the full “labelling → scoring → scenario → fallback” chain in _ct_decide_scenario.	Step-by-step verification confirms that each ticker correctly follows the full chain R_label/M_label/V_regime → scoring → scenario → final bucket. No divergence occurs between labels, selected scenario, and output. Fallback conditions remain inactive, indicating that rules fully cover observed situations.	VALIDATED

15 — Covid Crash (Extreme Volatility)

Data window	2019-09-03 → 2020-06-29
Frequency	Daily — 208 observations. Sufficient depth: short- and long-term indicators are statistically reliable. Warning: the observation window exceeds the typical interpretability range for this frequency (+4.0% above the recommended upper bound). Results may blend multiple market regimes, reducing economic signal clarity.
Tickers	QQQ, SPY, XLE, XLF, XLV
Indicators	Short term: 20 days volatility, 41 days return, 62 days momentum Long term: 50 days volatility, 82 days return, 186 days momentum

S15 — Covid Crash (Extreme volatility)

Test	Objective	Observation	Verdict
C8	Verify consistency between volatility-profile labels and “dynamic/speculative” comments.	The alignment between volatility-profile labels and generated comments is correct: each asset labelled “dynamic” or “speculative” receives a description consistent with its observed volatility level. No mismatches (e.g., speculative → calm) were detected.	VALIDATED
C10	Verify the full “labelling → scenario → fallback” chain in _ct_decide_scenario.	Full verification confirms correct generation of R/M/V labels and strict adherence to hierarchical rules during scenario selection. Compatible cases (e.g., vol_compress) are detected unambiguously, while others logically fall back to a neutral regime. No unexpected behavior or impossible scenario was observed.	VALIDATED
R2	Verify consistency of values and ratios across multiple volatility regimes.	Volatility values, ratios, and associated signals accurately reflect multi-regime Covid dynamics: high long-term volatility followed by rapid compression, sector-divergent momentum, and perfect coherence between ratios and historical price behavior.	VALIDATED
R6	Verify consistency between “subdued/normal/elevated/turbulent” labels and empirical variance.	“Subdued” and “normal” classifications align correctly with measured empirical variance. Volatility levels confirm a calm regime but remain normal relative to the post-crash Covid period. No mismatch between labels and statistical behavior was detected.	VALIDATED
R5	Verify that “stable/volatile” signals are coherent across noisy series.	“Stable/volatile” signals correctly reflect the multi-regime structure: noisy short-term series (post-crash) are labelled “subdued/normal,” while LT volatility remains elevated with logical consolidation momentum afterward. Labels match real statistical behavior across all assets.	VALIDATED
D1	Verify that calculated and labelled betas correspond to relative benchmark behavior.	Beta validation shows perfect correspondence between calculated values and assigned labels. Defensive sectors (XLV) are correctly labelled “defensive,” while SPY/XLF/XLE/QQQ fall under “market,” consistent with their actual behavior relative to the benchmark. No mismatch or anomalous threshold behavior was detected.	VALIDATED

16 — Inflation Spike (macro test)

Data window	2021-01-01 → 2022-12-01
Frequency	Monthly — 24 observations. Sufficient depth: short- and long-term indicators are statistically reliable.
Tickers	SPY, XLB, XLE, XLF, XLU
Indicators	Short term: 3 months volatility, 4 months return, 7 months momentum Long term: 7 months volatility, 8 months return, 21 months momentum

S16— Inflation Spike (macro test)

Test	Objective	Observation	Verdict
D1	Verify that calculated and labelled Betas correspond to relative benchmark behavior.	Beta values are consistent with observed dynamics: SPY serves as pivot at 1.0, cyclical sectors (XLE, XLF, XLB) show slightly lower sensitivity in line with their betas ~0.8, and XLU shows a dampened reaction confirming its defensive label. No conflict between behavior and labels was detected.	VALIDATED
D3	Verify bucket corrections via <code>_ct_adjust_bucket</code> on combined cases.	Bucket corrections follow the expected rules: low liquidity only downgrades high scenarios, defensive assets correctly shift high to moderate, and high-beta assets not present in this scenario do not trigger unintended severity escalations. Illiquid + defensive combinations such as XLU behaved as expected with no issues.	VALIDATED
R6	Verify consistency between “subdued/normal/elevated/turbulent” classifications and empirical variance.	Volatility regimes match empirical variance: tickers with ratios < 1 are correctly classified as subdued or normal based on their structural profiles, and no ratio > 1 is assigned. Sectors accustomed to higher volatility (XLE) are not mislabelled as elevated. No conflict between statistical behavior and labels was observed.	VALIDATED
R3	Verify that momentum and its ratio follow actual price dynamics.	Momentum dynamics precisely follow price movements for each ticker: upward accelerations → rising momentum, periods of stagnation → declining momentum. Ratio magnitudes match the amplitude of price changes, with no structural anomalies. Defensive (XLU) and cyclical (XLE) sectors behave consistently with their respective profiles.	VALIDATED
R4	Verify that RAR scores respect the logical Return/Volatility hierarchy.	Risk-Adjusted Returns respect the expected hierarchy: assets with the highest returns (XLE, XLF) display the highest RAR scores, while assets in decline (SPY, XLU) obtain negative RARs. With comparable volatility, relative ranking between XLB and XLF is consistent. No logical inversion was observed.	VALIDATED

17 — Cross-frequencies

Data window	2020-01-02 → 2024-10-21
Frequency	Cross frequencies: daily, weekly, monthly
Tickers	QQQ, SPY, XLE, XLF, XLK
Indicators	Diverse indicators

S17 — Cross-frequencies

Test	Objective	Observation	Verdict
D5	Verify that relative trends remain consistent across frequencies.	Relative trends observed across the three reports (daily, weekly, monthly) remain consistent: QQQ and SPY systematically dominate the dynamics, XLF follows an intermediate cyclical profile, XLE shows logical energy-inflation inertia, and XLK retains a more amplified high-beta tech profile. Directional rankings, momentum hierarchy, and volatility spread all remain structurally identical despite frequency changes. No trend inversion or inter-frequency anomaly was detected.	VALIDATED
C14	Verify that two identical reports differing by only one variable remain consistent.	We compared three reports identical in structure (profiles, metadata, sections) but differing only in frequency (daily / weekly / monthly). Frequency-independent components remained unchanged, while sensitive indicators (returns, momentum, CT scenarios) varied logically. No structural drift or cross-contamination was observed.	VALIDATED
D2	Verify consistency between raw variation and “strong/weak/neutral” labels.	Consistency between raw variation (returns, momentum, ratio) and qualitative labels was verified by systematically comparing numeric amplitudes with the applied classifications. Values remained coherent across all tested regimes, including extreme contexts. No mismatch between movement intensity and assigned label was found.	VALIDATED
C10	Verify the full “labelling → scenario → fallback” chain in _ct_decide_scenario.	The full labelling → scenario → fallback chain was validated across several frequencies and market regimes (Covid crash, inflation, high volatility). Generated labels feed correctly into _ct_decide_scenario, and prioritized rules consistently produce a coherent scenario. Fallback logic behaves as expected and no scenario yields an illogical or incomplete output.	VALIDATED

18 — Extreme Windows

Data window	2024-01-01 → 2024-02-01 & 2020-01-01 → 2024-10-22
Frequency	Daily — 145 observations. Sufficient depth: short- and long-term indicators are statistically reliable.
Tickers	AAPL, META, MSFT, TSLA
Indicators	Diverse indicators

S18 — Extreme Windows

Test	Objective	Observation	Verdict
L2	Verify the consistency of indicators across short, medium, and long windows.	Indicators were evaluated across short, medium, and long windows. On short windows, CT signals remain directionally coherent despite noise; on medium windows, CT and LT stay aligned; on long windows, LT smooths appropriately while CT remains stable. No structural artefact or inconsistency was observed.	VALIDATED
R2	Verify the consistency of values and ratios across multiple volatility regimes.	Volatility_Short / Volatility_Long ratios were checked across several assets with different noise levels. Calm assets display ratios near 1, while noisier assets show ratios below 1, as expected from their dynamic profile. Regimes (subdued / normal) align perfectly with these volatility levels, validating R2.	VALIDATED
R5	Verify that “stable / volatile” signals are coherent across noisy series.	“Stable / volatile” signals were compared across assets with low noise (AAPL/MSFT), moderate noise (META), and high noise (TSLA), across short and long windows. CT/LT ratios correctly reflect: < 1 for smooth assets, ≈ 1 for moderately noisy assets, and > 1 for the noisiest assets. Assigned regimes (subdued / normal) match real statistical behavior, validating R5.	VALIDATED
C14	Verify that two identical reports differing by only one variable remain consistent.	Two reports generated under the same settings and structure but differing only in window length were compared. All window-independent components remained identical. Only variable-dependent sections (CT/LT metrics, depth of history, and return/momentum regimes) changed logically. No drift or unexpected discrepancy was observed across blocks, validating C14.	VALIDATED

19 — Visualisation / Graph Matching

Data window	2022-01-03 → 2024-10-21
Frequency	Weekly — 147 observations. Sufficient depth: short- and long-term indicators are statistically reliable. Warning: the observation window exceeds the typical interpretability range for this frequency (+182.7% above the recommended upper bound). Results may blend multiple market regimes, reducing economic signal clarity.
Tickers	AAPL, JNJ, JPM, MSFT, NVDA
Indicators	Short term: 8 weeks volatility, 12 weeks return, 16 weeks momentum Long term: 20 weeks volatility, 24 weeks return, 48 weeks momentum

S19— Visualisation / Graph Matching

Test	Objective	Observation	Verdict
C11	Verify the rendering of aggregated text: ordering, punctuation, and PDF formatting.	Text rendering + aggregation + PDF export works perfectly, with no artefacts, no truncation, no interpolation issues, and no structural problems.	VALIDATED
C13	Verify that values displayed in charts strictly match the data in the tables.	The test confirms that all chart series (price, returns, momentum, volatility) are perfectly aligned with the values shown in the report tables. Transformations applied (base-100 indexation, rolling means, LT/ST windows) match exactly the methods used to compute indicators. No divergence or offset was detected between tabular and graphical data. Inter-frequency consistency (daily/weekly/monthly) confirms the robustness of the pipeline.	VALIDATED

20 — Correlation Matrix Validation

Data window	2020-01-01 → 2024-10-22
Frequency	Weekly — 252 observations. Sufficient depth: short- and long-term indicators are statistically reliable. Warning: the observation window exceeds the typical interpretability range for this frequency (+384.6% above the recommended upper bound). Results may blend multiple market regimes, reducing economic signal clarity.
Tickers	GLD, QQQ, SPY, TLT, VNQ, XLE
Indicators	Short term: 8 weeks volatility, 12 weeks return, 16 weeks momentum Long term: 20 weeks volatility, 24 weeks return, 48 weeks momentum

S20— Correlation Matrix Validation

Test	Objective	Observation	Verdict
D4	Verify that strong/anti-correlated pairs are correctly represented on the matrix.	I compared correlation coefficients from the table with the graphical matrix, verifying consistency between extreme values (> 0.7) and their visual representation. Structurally correlated pairs (SPY–QQQ–VNQ) appear in warm zones, while anti-correlated assets (TLT vs equities, GLD vs equities, energy vs tech) clearly appear in cold zones. No inversion of sign or numerical inconsistency was detected, confirming that the correlation engine correctly reflects economic relationships.	VALIDATED
C13	Verify that values displayed in charts strictly match the data in the tables.	I compared all key values used for chart construction (returns, momentum, volatility, base-100 series) with those in the report tables, checking alignment, temporal structure, peaks, local variations, and extremums. Visual inflection points match precisely with changes in trend, ratios, and levels found in the tables. No offset, excessive smoothing, or undocumented transformation was detected.	VALIDATED
C9	Verify that final concatenation produces a coherent, contradiction-free narrative.	I verified that the concatenation of blocks (volatility profiles, labels, scenarios, interpretations, and metrics) produces a coherent narrative without contradictions: no label conflicts with comments, and all interpretations align with R/M/V labels used upstream. Aggregated sections maintain logical continuity across tables, explanatory paragraphs, and scenario buckets.	VALIDATED

21 — Japan market

Data window	2018-12-31 → 2024-10-21
Frequency	Weekly — 304 observations. Sufficient depth: short- and long-term indicators are statistically reliable. Warning: the observation window exceeds the typical interpretability range for this frequency (+484.6% above the recommended upper bound). Results may blend multiple market regimes, reducing economic signal clarity.
Tickers	6758.T, 7203.T, 8035.T, 8306.T, 9432.T
Indicators	Short term: 8 weeks volatility, 12 weeks return, 16 weeks momentum Long term: 20 weeks volatility, 24 weeks return, 48 weeks momentum

S21 — Japan market

Test	Objective	Observation	Verdict
C2	Verify the consistency of “Partial coverage” or “Missing sectors” messages according to missing GICS sectors.	For C2, the engine correctly identifies that no GICS sector is actually represented in the portfolio. The messages “Partial coverage” and “Significant sector gaps” appear automatically without contradiction. This behavior confirms that the engine correctly flags lack of sector diversification when the data justify it.	VALIDATED
C3	Verify that detected biases (growth-oriented, defensive bias, etc.) correspond to the sectors present.	For C3, a detailed review of the report comments shows that the “growth-oriented” bias is coherent with the portfolio structure: two tickers (6758.T and 8035.T) concentrate most of the factor and return risk, clearly exhibiting a long-duration and growth profile. Other tickers, even defensive or value-leaning ones, do not offset this dominant signal.	VALIDATED
C9	Verify that final concatenation produces a coherent view with no contradictions.	I compared profiles, labels, scenarios, correlations, and macro comments from report S20. All sections describe the same structure: moderate volatility, defensive bias, concentration risk, and absence of broad sector coverage. No internal conflict (e.g., volatility, beta, liquidity) appears between tables, charts, or aggregated text. The final narrative is coherent and contradiction-free.	VALIDATED
R1	Verify the accuracy of return calculations and their graphical coherence.	Short-term return values and AReturn displayed in the report fully match the R1 curves. For each ticker, direction, amplitude, and sign of CT returns correspond exactly to visible movements near the end of the series. Rising assets show modest positive returns, and declining assets show slightly negative returns. CT–LT differences correctly reflect real dynamics. Overall, return calculations are validated.	VALIDATED
D1	Verify that calculated and labelled Betas correspond to relative benchmark behavior.	Calculated Betas for Japanese equities accurately reflect their sensitivity relative to the benchmark: defensive tickers (Toyota, NTT, MUFG) show stable Betas, Sony displays a slightly higher Beta consistent with market behavior, and Tokyo Electron shows a Beta > 1 coherent with its cyclical tech profile. No mismatch was detected.	VALIDATED

22 — German Market

Data window	2022-01-01 → 2025-10-01
Frequency	Monthly — 46 observations. Sufficient depth: short- and long-term indicators are statistically reliable.
Tickers	BAYN.DE, DTE.DE, HEN3.DE, LIN.DE, SAP.DE
Indicators	Short term: 4 months volatility, 9 months return, 12 months momentum Long term: 10 months volatility, 18 months return, 36 months momentum

S22 — German market

Test	Objective	Observation	Verdict
C2	Verify the consistency of “Partial coverage” or “Missing sectors” messages according to missing GICS sectors.	I compared the GICS sectors of the five equities against the 11 standard sectors. Several major sectors are missing, which objectively constitutes partial coverage. The report correctly identifies this lack of sector diversity and does not falsely signal the presence of absent sectors. Consistency between GICS data and the produced message is validated.	VALIDATED
C3	Verify that detected biases (growth-oriented, defensive bias, etc.) correspond to the sectors present.	I verified Betas, Defensive/Market labels, and associated sectors. The portfolio is dominated by defensive, low-beta stocks with high growth exposure. The sector bias detected by the engine corresponds exactly to the sectors actually present. Consistency is validated.	VALIDATED
C4	Check the logic for “Large/mega-cap tilt” / “Small/micro-cap tilt” / “Barbell by size.”	I verified Market_Cap and the associated labels (large/mega only). The absence of small-caps makes a barbell or small/micro tilt impossible. The engine correctly outputs a pure large/mega tilt, fully consistent with the portfolio composition.	VALIDATED
R1	Verify the accuracy of return calculations and their graphical consistency.	The short-term return curve reacts exactly to directional variations visible in price. For SAP.DE, the rise in 2023–2024 results in sustained positive returns, while the 2025 collapse immediately produces strongly negative returns (coherent amplitude, ~-30%/-40% annualized). The same applies to DTE.DE and BAYN.DE: each price spike produces the corresponding return spike, and each rebound is captured in CT returns. No structural divergence was observed, confirming that CT returns are derived directly from price variations across the sliding window.	VALIDATED
R3	Verify that momentum and its ratio follow actual price dynamics.	Long-term momentum follows the general slope of price across the full window. For SAP.DE, momentum increases gradually during the 2022–2024 uptrend, then declines sharply in the 2025 reversal. For BAYN.DE, the prolonged downward trend results in consistently negative and falling momentum. M/M_LT ratios increase during acceleration phases (e.g., LIN.DE mid-2024) and fall when CT momentum drops faster than LT. Observed values perfectly match long-term dynamics.	VALIDATED
R7	Verify that the ADV10USD median aligns with displayed liquidity profiles.	I compared each ticker’s percentile in the internal thin/illiquid matrix. Very low values (0.015M, 0.38M) are correctly classified as illiquid. Medium values (~1.2M) fall logically into the thin/illiquid category. BAYN.DE (>4M) remains within the moderate zone but is consistent with European liquidity norms. Labels fully reflect empirical reality.	VALIDATED

23 — Compressed Volatility

Data window	2020-12-31 → 2021-12-31
Frequency	Daily — 253 observations. Sufficient depth: short- and long-term indicators are statistically reliable. Warning: the observation window exceeds the typical interpretability range for this frequency (+26.5% above the recommended upper bound). Results may blend multiple market regimes, reducing economic signal clarity.
Tickers	QQQ, SPY, XLK, XLP, XLU
Indicators	Short term: 25 days volatility, 50 days return, 75 days momentum Long term: 62 days volatility, 100 days return, 225 days momentum

S23 — Compressed Volatility

Test	Objective	Observation	Verdict
C8	Verify consistency between volatility-profile labels and “dynamic/speculative” comments.	Consistency between volatility-profile labels and wording is correct: when an asset is dynamic, the comment adopts terms like “market-driven, signal unclear.” No asset receives an inappropriate “speculative” label since none matches that profile. No mismatch in technical content or wording was detected.	VALIDATED
R6	Verify consistency between “subdued/normal/elevated/turbulent” categories and empirical variance.	Categories are perfectly aligned with empirical variance: low-vol assets are tagged “subdued,” intermediate volatilities are labelled “normal,” and no asset is incorrectly classified as elevated/turbulent. The low short/long ratio (<1 for all) confirms a calm environment. Classification is judged coherent.	VALIDATED
R2	Verify consistency of values and ratios across multiple volatility regimes.	Ratios accurately describe the environment: volatility is expanding while momentum remains low, which corresponds exactly to the wording “volatility expanding / momentum stable.” Low AReturn values reinforce the coherence of the declared “neutral” scenario. No ratio contradicts the observed regime.	VALIDATED
R5	Verify that “stable/volatile” signals are consistent across series with different noise levels.	Signals precisely reflect noise level: low-variance series are labelled “neutral/stable,” and no excessive signal appears where volatility is minimal. The engine does not overinterpret micro-variation and remains robust on calm series. Behavior is judged coherent.	VALIDATED

24 — Crypto Vol Spike

Data window	2021-05-01 → 2021-07-30
Frequency	Daily — 91 observations. Sufficient depth: short- and long-term indicators are statistically reliable.
Tickers	ADA-USD, BTC-USD, DOGE-USD, ETH-USD
Indicators	Short term: 9 days volatility, 18 days return, 27 days momentum Long term: 22 days volatility, 36 days return, 81 days momentum

S24 — Crypto Vol Spike (mini-fenêtre)

Test	Objective	Observation	Verdict
R2	Verify consistency of values and ratios across multiple volatility regimes.	I checked that assets exhibit differentiated volatility regimes (e.g., BTC long vol 2.12% vs ADA long vol 3.39% vs DOGE 4.32%) and that Volatility_Ratio (CT vs LT) correctly reflects compression states. AReturn and Momentum_Ratio are coherent with price/momentum evolution shown in charts (BTC accelerates while ADA/DOGE show stagnation or reversal).	VALIDATED
R5	Verify that “stable/volatile” signals are consistent across noisy series.	I compared smoother series (BTC, ETH) with noisier ones (DOGE, ADA). Signal labels (“weak reversal,” “neutral,” “mildly coherent”) vary exactly according to noise level and directional quality. DOGE and ADA, being erratic, correctly receive weak signals; BTC/ETH, more directional, obtain more coherent signals.	VALIDATED
C10	Verify the full “labelling → scenario → fallback” chain in _ct_decide_scenario.	The full chain works without collision, misprioritization, or unresolved paths. All labels flow correctly into scenario selection and fallback rules behave as expected.	VALIDATED
R8	Verify that indicators remain stable/reasonable in the presence of jump events (earnings, flash crash).	I examined Momentum(%) and Momentum_LT(%): no anomaly appeared, including typical spike artifacts (e.g., momentum ratio >10 or <-10). Filtering correctly absorbs sudden jumps and indicators remain within a reasonable range (Momentum_Ratio between 0.27 and 1.03; AReturn between 0.13 and 0.86). No extreme scenario was triggered erroneously.	VALIDATED

25 — Full-Scale Validation (global mix)

Data window	2019-12-30 → 2024-10-21
Frequency	Weekly — 252 observations. Sufficient depth: short- and long-term indicators are statistically reliable. Warning: the observation window exceeds the typical interpretability range for this frequency (+384.6% above the recommended upper bound). Results may blend multiple market regimes, reducing economic signal clarity.
Tickers	6758.T, ARE, BTC-USD, CADJPY, COST, EURCAD, LOW, LVS, MRNA, PSCH SAN.PA, SAP.DE, SPXL, TLT, XLU
Indicators	Short term: 8 weeks volatility, 12 weeks return, 16 weeks momentum Long term: 20 weeks volatility, 24 weeks return, 48 weeks momentum

S26 — Full-Scale Validation (global mix)

Test	Objective	Observation	Verdict
C9	Verify that final concatenation produces a coherent, contradiction-free narrative.	For this test, I compared the three blocks: profiles → labels → PDF interpretations. The engine shows no internal contradiction: an asset classified as speculative in the profile (ARE, LVS, SPXL) retains this status in the KPI table and in the “Current Regime Snapshot.” Likewise, calculated metrics (AReturn, Momentum Ratio, Vol Ratio) are reused in the PDF comments without alteration.	VALIDATED
R6	Verify consistency between “subdued/normal/elevated/turbulent” categories and empirical variance.	I checked each asset by comparing Volatility_Short, Volatility_Long, and Volatility_Profile. No inconsistency was found: the engine strictly follows empirical logic.	VALIDATED
D3	Verify bucket corrections via _ct_adjust_bucket on combined cases.	Results from the KPI table and the Stress table confirm that bucket adjustments were applied correctly.	VALIDATED
R4	Verify that RAR scores respect the logical Return/Volatility hierarchy.	To validate this test, I compared RAR (Return / Volatility — internal logic of the engine) with the values in the KPI table. All RAR scores follow the expected ranking: no underperforming but highly volatile asset receives an inflated RAR.	VALIDATED
L1	Verify visual stability and readability of pages for universes of 10 to 60 tickers.	I inspected the structure of pages: tables, heatmaps, charts, macro sections. Layout remains stable. However, variable table lengths impose spacing conditions between sections.	VALIDATED
L3	Verify engine robustness on recent or sparsely traded tickers.	This scenario includes critical cases: PSCH (adv 16.5K), EURCAD (forex without Market Cap), CADJPY, ARE (very high volatility). Metrics remain calculated, labels remain coherent, and PDF sections do not crash despite missing capitalization or ADV fields.	VALIDATED
C13	Verify that values displayed in charts strictly match the data in tables.	No divergence or inconsistency was found between tables and charts.	VALIDATED
R9	Verify engine stability in the presence of missing, erroneous, or inconsistent data.	The PDF does not crash; missing fields are replaced properly (“—”), and any undefined variable (risk, momentum, regime) does not generate “NaN.”	VALIDATED

26 — Mix Region

Data window	2019-12-30 → 2024-10-21
Frequency	Weekly — 252 observations. Sufficient depth: short- and long-term indicators are statistically reliable. Warning: the observation window exceeds the typical interpretability range for this frequency (+384.6% above the recommended upper bound). Results may blend multiple market regimes, reducing economic signal clarity.
Tickers	6758.T, 7203.T, ORA.PA, QQQ, SAP.DE, SPY
Indicators	Short term: 8 weeks volatility, 12 weeks return, 16 weeks momentum Long term: 20 weeks volatility, 24 weeks return, 48 weeks momentum

S27— Mix Region

Test	Objective	Observation	Verdict
D3	Verify bucket corrections via <code>_ct_adjust_bucket</code> on combined cases.	I checked each ticker combining illiquid/thin, high beta, and leveraged profiles. None required additional downgrades beyond rule-based adjustments, and final buckets do not contradict any <code>_ct_adjust_bucket</code> condition. Coherence between rules and output is fully confirmed.	VALIDATED
R7	Verify that the ADV10USD median aligns with the displayed liquidity profiles.	I compared individual ADV10 values with the group median and the engine's thresholds. Tickers >50M are classified "liquid," those around 10–20M "thin," and those near 1–4M "illiquid." Labels accurately match the observed distribution.	VALIDATED
R9	Verify engine stability in the presence of missing, erroneous, or inconsistent data.	I examined tickers without valid Market Cap or ADV (e.g., forex). The engine produces values and labels without crashing; missing fields are handled correctly, and metrics not applicable are ignored. Stability remains intact despite partial data.	VALIDATED
C9	Verify that final concatenation produces a coherent, contradiction-free narrative.	I compared all PDF sections (profiles, KPIs, rotation, macro). Labels, comments, and numerical results remain aligned with no internal contradiction. No inversion of meaning or narrative inconsistency appears in any block.	VALIDATED
L3	Verify engine robustness on recent or thinly traded tickers.	I observed SAP.DE, ORA.PA and other thin/illiquid tickers. Return, momentum, and volatility calculations remain stable and no pipeline error appears. Generated labels remain coherent despite limited liquidity.	VALIDATED